

Technology Stack

Date	15 March 2026
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit	Chosen to build a simple Python-based web interface with concept input, audio upload, playback, result display, waveform viewing, and PDF download features.

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2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python	Used to handle AI transcription, semantic similarity, audio analysis, filler detection, scoring logic, and PDF report generation.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	No Database / Local File Storage	The project does not use a database. Uploaded audio files, waveform images, and generated PDF reports are stored temporarily in local folders during runtime.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render / VPS / Local Streamlit	Supports local demonstration and online hosting. VPS or Render can run the Streamlit application with FFmpeg and required AI dependencies.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git and GitHub	Used for version control, source-code backup, collaboration, and deployment integration.

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6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenAI Whisper, Sentence-BERT, Librosa, ReportLab, imageio-ffmpeg	Whisper converts speech to text, Sentence-BERT compares concept meaning, Librosa extracts audio features, ReportLab creates PDF reports, and imageio-ffmpeg supports audio processing.